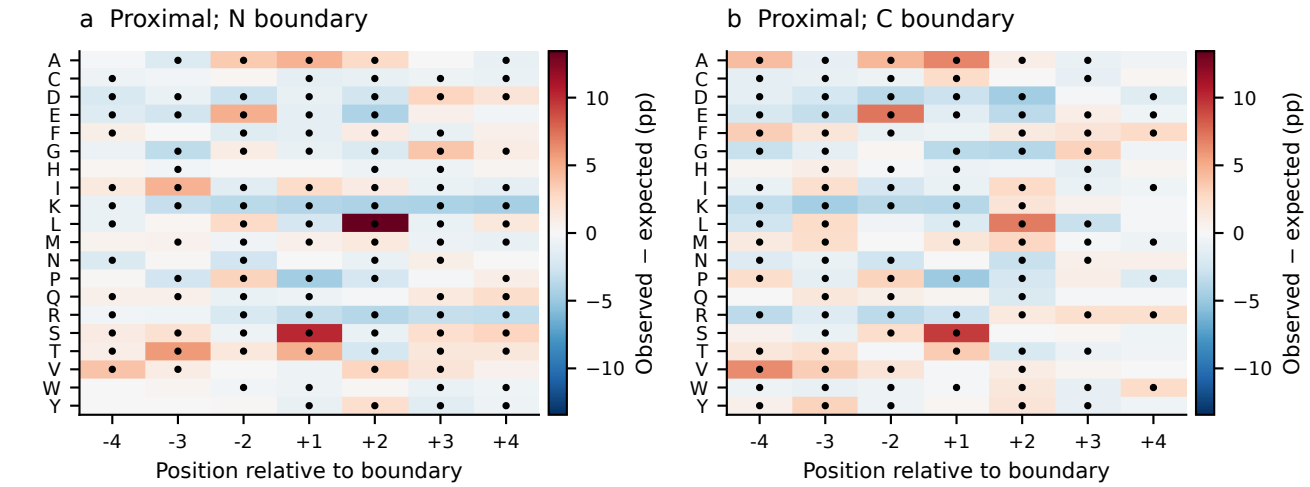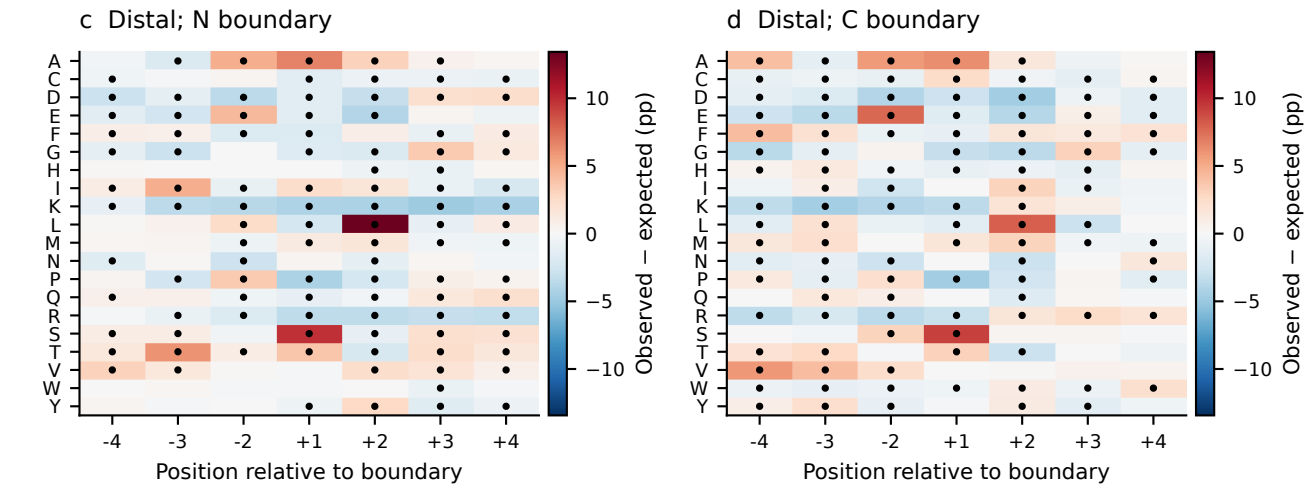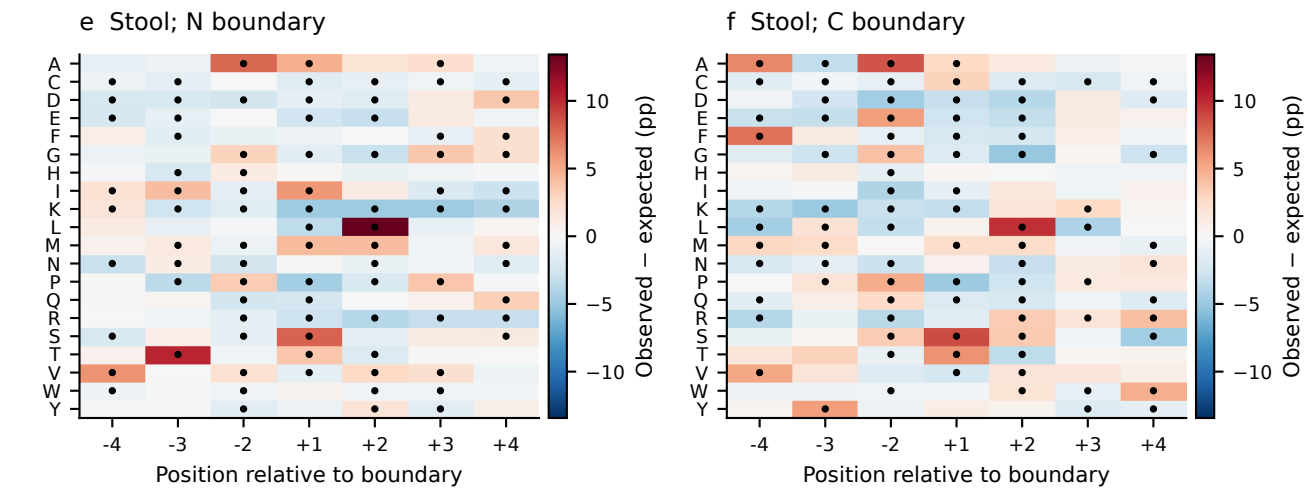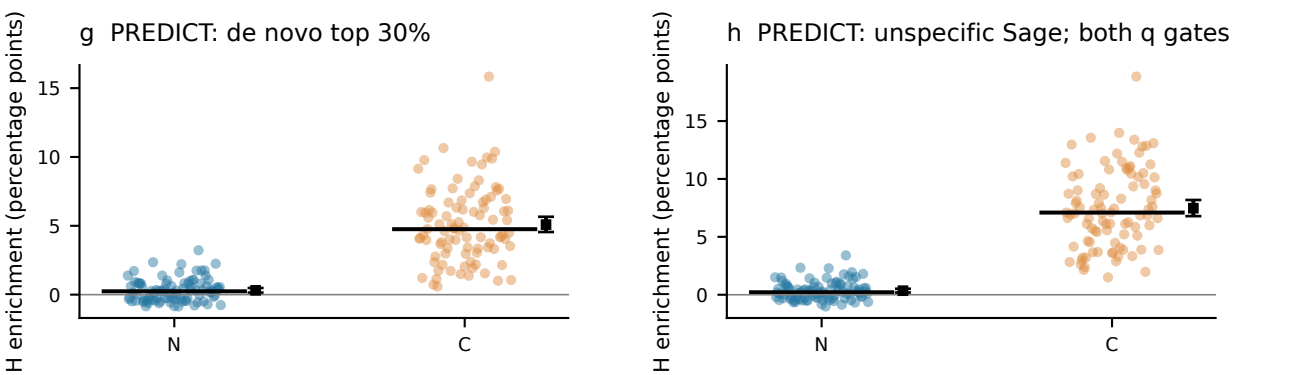Histidine context and replication across cohorts